

HOT FORMED HOLLOW SECTIONS

IMPACT TEST

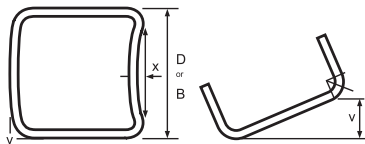
Standard impact specimens are cut longitudinally from the sections and machined on each face to 10mm x 10mm in cross section and a V notch, 2 mm deep, is cut into one face. The specimen is then tested at the appropriate temperature in accordance with BS 131: Part 2, The Charpy V-Notch Impact Test. Due to the machining, standard specimens can only be taken from material 11 mm thick, or thicker. For thinner material, tests are made on the next subsidiary standard test piece with average values in accordance with Table 15 of BS 4360.

Average value for standard test piece	Equivalent average value for subsidiary standard test piece	
10mm x 10mm	10mm x 7.5mm	10 mm x 5.0 mm
J	J	J
27	22	19

Unless agreed otherwise, impact tests are not carried out on Grade 43C material or where the section is less than 6 mm thick.

DIMENSIONAL TOLERANCES

The dimensional tolerances are in accordance with BS 4848, Part 2, as follows:



(a) Circular Hollow Sections Outside diameter	$\pm 0.5\text{mm}$ or $\pm 1\%$ whichever is the greater
(b) Rectangular Hollow Sections	
(i) Outside dimensions of sides	$\pm 0.5\text{mm}$ or $\pm 1\%$ whichever is the greater
(ii) Squareness of side	$90^\circ \pm 1^\circ$
(iii) Radii of corners	Outside - between the limits of $0.5t$ and $2.0t$ Inside - between the limits of $0.5t$ and $1.5t$ where t equals the specified thickness of section
(iv) Concavity/Convexity (x)	$\pm 1\%$ of the length of the side, (this tolerance measured independently of the tolerance on outside dimension.)
(v) Angular twist (v)	2 mm plus 0.5 mm per metre maximum. Twist is measured by laying the section, as produced, on a horizontal surface with the face at one end pressed flat. The difference in height above the surface between the two corners at the opposite end is measured.

MASS TOLERANCE

The rolling tolerance in accordance with BS 4848, Part 2 is $\pm 6\%$ on individual lengths; + 6% - 4% on lots of 10 tonnes and over.

STRAIGHTNESS TOLERANCE

Unless otherwise arranged, the hollow section shall not deviate from straightness by more than 0.2% of the total length as produced, measured at the centre of the length.

LENGTH TOLERANCE

Mill and Random lengths are supplied as in the following tables. Exacts may be supplied at our option, with a tolerance of +6 mm - 0. Lengths above 12 m in length are subject to an extra for transportation.